

WORK REPORT

Project Lead – On-Grid Solar Panel System Installation

Name	Mohammed Siddiq M S
Role	Project Lead
Project Title	On-Grid Solar Panel System with Net Metering and Excess Energy Export to EB Grid
Duration	January 2022 – March 2023
Institution	Rohini College of Engineering and Technology
Recognition	First Runner-Up, Smart Minds Hackathon (National Level Event), organized by G20, NITI Aayog and AIC RAISE, 24th-25th March 2023

1. Project Overview

Between January 2022 and March 2023, I worked on a project focused on the design and implementation of an On-Grid Solar Panel System. The objective of the project was to develop a renewable energy solution that not only meets the household's electricity requirements but also exports the surplus energy generated to the Electricity Board (EB) grid through net metering. Under this arrangement, any excess power produced beyond the household's consumption is fed back into the grid, and the producer is compensated by the government/electricity board for the units supplied.

As the Project Lead, I was responsible for coordinating the team, planning the system design, overseeing implementation, and presenting the solution at a national-level platform.

2. Role and Responsibilities

As Project Lead, my key responsibilities included:

- Leading a team of student engineers in designing the on-grid solar power system.
- Coordinating the technical planning for solar panel sizing, inverter selection, and grid synchronization.
- Overseeing the net-metering concept, where excess solar energy generated is exported back to the EB grid and the household receives credit/payment for the surplus units.

- Managing project timelines, task allocation, and team communication throughout the development cycle.
- Preparing technical documentation and presentation material for evaluation.
- Representing the team in presenting the project to judges and stakeholders at the Smart Minds Hackathon.

3. Technical Description: On-Grid Solar Panel System

An on-grid (grid-tied) solar panel system is connected directly to the public electricity grid. Solar panels installed on the rooftop convert sunlight into DC electricity, which is then converted to AC electricity using a grid-tied inverter. This AC electricity is used to power household appliances first.

The key feature of this project was the net-metering mechanism:

- During daylight hours, the solar system generates electricity that is consumed by the household.
- If the solar generation exceeds the household's consumption, the surplus electricity is automatically exported to the EB grid through the net meter.
- The net meter records the units exported to the grid versus the units consumed from the grid.
- At billing cycles, the household is credited or paid by the electricity board for the net surplus units supplied to the grid, in accordance with the applicable net-metering policy.
- This approach reduces electricity bills for the consumer while contributing clean energy to the overall grid.

4. Recognition and Achievement

This project was presented at the Smart Minds Hackathon, a National Level Event organized by G20, NITI Aayog, and the Atal Incubation Center (AIC RAISE), held on 24th and 25th March 2023 at Coimbatore.

The team was awarded the First Runner-Up position in recognition of innovative ideas, effective communication, and strong teamwork in the areas of leadership, problem-solving, and business decision-making during the hackathon.

5. Skills and Competencies Gained

- Project leadership and team coordination
- Renewable energy system design (solar PV systems)
- Understanding of grid-tied inverter systems and net metering
- Technical presentation and documentation
- Problem-solving and business decision-making in a competitive environment

Annexure: Supporting Documents and Photographs

A. Award Ceremony Photograph

Photograph from the Smart Minds Hackathon award ceremony, AIC RAISE, Coimbatore, where the team received the First Runner-Up award.



B. AIC RAISE Logo (Organizing Body)

Official logo of AIC RAISE, supported by the Atal Innovation Mission, NITI Aayog, Government of India, the organization that conducted the Smart Minds Hackathon.



C. Certificate of Achievement

Certificate of Achievement issued by AIC Raise Startup Incubation Centre Rathinam Techzone, Pollachi Main Road Eachanari, Coimbatore, Tamil Nadu, India - 641021, confirming the Runner-Up award in the Smart Minds Hackathon.

